

THE INTERN DIARIES

Intern Report

Region Performance

WEEK 3 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

SuperStore (RETAIL COMMERCE)

1. How do I build a region summary in Excel that compares *both* sales and profit and ranks the regions?

Answer: On a Region_Summary tab, list the four regions in A3:A6 (exact match to the data) and drive every column off the Superstore table with structured references:

- Total Sales → =SUMIFS(Superstore[Sales], Superstore[Region], \$A4)
- Total Profit → =SUMIFS(Superstore[Profit], Superstore[Region], \$A4)
- Profit Margin → =IFERROR(C4/B4,"")
- Order Lines → =COUNTIFS(Superstore[Region], \$A4)
- Avg Profit / Line → =IFERROR(C4/E4,"")
- Sales / Profit / Margin Rank → =RANK.EQ(value, range, 0) (descending, rank 1 = biggest)
- Composite Score → =AVERAGE(salesRank, profitRank, marginRank), then Final Rank → =RANK.EQ(composite, range, 1) (**ascending**, because for an average-of-ranks a *lower* score is better).

2. Why is ranking regions on sales alone misleading — and what's the concrete example in this data?

Answer: Sales measures *scale*; profit and margin measure *business health*, and the two disagree exactly where it matters. In this dataset **Central** ranks 3rd on sales (\$501K) but last on both profit (\$40K) and margin (7.9%) — a sales-only leaderboard would call it a solid mid-tier region when it is in fact the weakest. Conversely **South** ranks last on sales (\$392K) yet beats Central on profit *and* margin (11.9% vs 7.9%) — smaller but healthier. The composite rank captures this correctly (South 3rd, Central 4th), which a single-metric rank would not.

“sales for scale, profit for value, margin for efficiency.”

3. How do I prove the region summary didn't silently drop or double-count any order lines?

Answer: Add a totals/control row (row 8) and three reconciliation checks (rows 10–12) that compare the breakdown back to the whole table, using a tolerance for money:

- `=IF(ABS(B8-SUM(Superstore[Sales]))<0.01," ✓ MATCH"," ✗ CHECK")`
- `=IF(ABS(C8-SUM(Superstore[Profit]))<0.01," ✓ MATCH"," ✗ CHECK")`
- `=IF(E8=COUNTA(Superstore[Region])," ✓ MATCH"," ✗ CHECK")` (must equal 9,994)

If any check fails, the cause is almost always a **label mismatch** (a region name with a trailing space or wrong case, or a blank region row) — fix the *label*, not the formula. This is the same control-total discipline that caught the truncated table on Day 16.

4. Which chart type fits "compare sales across regions," and why not a line or a pie?

Answer: A **clustered column** chart from A3:C6 (Region on the X-axis, Sales and Profit as two side-by-side bars), titled Sales vs Profit by Region, with axis labels and a legend. Regions are **discrete groups**, so column/bar is correct (the Day-7 chart-type rule): a **line** chart implies a time/ordered axis that regions don't have, and a **pie** can only show one measure's share, not a sales-vs-profit comparison.

5. How would you rank regions in a balanced, defensible way?

Answer: Not on one metric. The recommended approach is a **three-lens rank** with a composite:

- **Primary = Total Profit** (closest to business value),
- **Secondary = Total Sales** (market size),
- **Guardrail = Profit Margin** (efficiency / discount health),
- combined into a **composite score** (here the average of the three ranks, lower = better) and a final rank.